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FOG FORECASTING

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Fog is a major nuisance to travelers. Whether driving or flying, fog results in travel delays and in some cases cancellations. Forecasting fog can be difficult, but its proper prediction is extremely important. The proper prediction of fog can have people better prepared to avoid delays and being late for work. The best preparation is to leave early for work or school. Leaving early will help avoid the larger traffic jams that can result in urban areas on those foggy mornings.

The following is a list of the primary processes that produce fog. A combination of several of these factors increases the likelihood of fog:

1. Saturated air at surface

Fog forms once evaporation into the air results in supersaturation. If the dewpoint depression is small after sunset, clear skies will cause radiational cooling of the air quickly toward saturation. The dewpoint can increase due to a rain shower, previously saturated soils and irrigation. Since vegetation evapotranspires moisture, fog first forms over grassy and vegetated areas. Fog is common in situations where a daytime shower saturates the soil, vegetation and boundary layer and then skies clear in the evening into the night hours.

2. Overnight clear skies

Clear skies allow the surface temperature to cool off at a higher rate. If dewpoints are high or the dewpoint depression is low, saturation of the air will occur over night. Fog is not as likely if the ground is bone dry, vegetation is sparse or the dewpoint depression is too large.

3. Wet soils and rain dampened vegetation

Wet soils and dampened vegetation continuously evaporates water vapor. This allows the temperature and the dewpoint to converge more rapidly than would normally be the case. An afternoon shower can cause the dewpoint depression to be near zero in the evening. If skies then clear and wind is light, fog is very likely.

4. Light wind

Fog requires a mixing action by wind; without wind, dew will appear instead of fog. If the

surface is near saturation, a light wind will allow for the layer of air near the surface to remain near saturation. High wind speeds cause a mixing of air at the surface and higher into the atmosphere. Since air higher in the atmosphere is generally drier, high wind dries the air and prevents fog from forming.

5. Slight warm air advection from maritime polar/ tropical air

Warm air advection causes rising air. Even a slight warm air advection can cause just enough uplift to make fog more likely, especially if the warm air advection is from a moisture source such as the Gulf of Mexico, Great lakes or Gulf Stream. Polar air and continental air tend to have larger dewpoint depressions than maritime air. Fog that occurs in polar and continental air is primarily due to saturation from above or large radiational cooling.

6. High dewpoint

Warm air has a higher capacity to contain and evaporate water vapor than cold air. Because of this, fog associated with maritime air tends to be thicker than fogs that form at very low temperatures.

7. Light drizzle and precipitation saturating PBL from above

Warm moist air overrunning a shallow airmass can saturate the shallow air from above and eventually to the surface. The contact cooling between the two air masses causes clouds and fog since the moisture in the warm air is beyond the carrying capacity of moisture relative to the cooler air. Precipitation evaporating into the air can saturate the atmosphere.

8. Wind direction from a moisture source

Since moisture is a key component for fog, advection from a moisture source is much more favorable than advection from a dry source. Widespread fog is more common with warm fronts than cold fronts. Part of the reason is that cold fronts tend to bring continental air (central and eastern US) while a south wind brings maritime warm air.

9. Upslope flow

Rising air lowers the dewpoint depression.

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OUTLINE/RULES OF THUMB FOR FOG DEVELOPMENT/DISSIPATION

Fog is formed when air at or near the earth's surface becomes saturated by any of the three processes - cooling, addition of moisture, or mixing with another air parcel. Generally, fog forms in a stable air mass environment. Fog does not generally form with an unstable atmosphere because vertical mixing results in convective or layered cloudiness.

Fog classifications:

1. Radiational
 - A. Ground fog
 - B. Continental high inversion fog
2. Advection (sea or lake fog)
3. Combination
 - A. Upslope fog
 - B. Advection-radiation fog
 - C. Frontal fog

The cooling process may involve one or more of the following:

1. Nocturnal radiational cooling of the earth's surface, which results in cooling the lowest layer of air near the surface.
2. Advection of air over a colder surface.
3. Adiabatic cooling due to lifting an air parcel either by force lifting or orographic lifting.
4. Evaporative cooling; precipitation falling through drier air.

The addition of moisture to an air mass may be brought about by one or more of the following:

1. Advection of moisture.
2. Evaporation from a wet surface.
3. Evaporation from falling precipitation.
4. Turbulent mixing of moisture in the lower portion of the atmosphere.

General rules of thumb for fog forecasting:

1. If the mixing ratio increases with height in the boundary layer, consider forecasting fog. This assumes clear skies and no dry air advection.
2. If the mixing ratio decreases with height in the boundary layer, forecast dew or frost (assuming clear skies and no moisture advection).
3. If rain ends late in the day, followed by clearing and little wind, consider forecasting fog.

Fog/stratus dissipation - removing moisture and/or heating the air

1. Moisture is decreased by the following:
 - A. Turbulent transfer of moisture downward to the surface (e.g., to form dew or frost).
 - B. Turbulent mixing of the fog layer with adjacent drier air.
 - C. Advection of drier air.
 - D. Condensation of the water vapor to clouds.
2. Heating of the air results from the following:
 - A. Turbulent transport of heat upward from air in contact with warm ground.
 - A. Advection of warmer air.
 - B. Transport of the air over a warmer land surface.
 - C. Adiabatic warming of the air through subsidence or downslope motion.

- D. Turbulent mixing of the fog layer with adjacent warmer air aloft.
- E. Release of latent heat associated with the formation of clouds.

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DISSIPATION OF RADIATION ROG

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The dissipation of radiation fog can take place through the following mechanisms:

- The spreading of a cloud sheet above the fog, which reduces the radiative cooling at the fog top.
- Atmospheric layers with high amounts of water vapour can help to dissipate fog by long wave radiation.
- Heat from the ground can lift the fog into low cloud.
- Increasing shear at fog top caused by increasing wind above the Fog entrains drier air into the cloud and accelerates the dissipation.
- In mature Fogs the settling of droplets to the ground or onto vegetation contributes significantly to the dissipation.
- Solar radiation (excluding mid-winter) can penetrate to the ground in places where Fog is thinner (such as elevated places) and heat the ground and evaporate the Fog. Consequently, heat will be convected into the cloud. During summertime due to short nights the Fog has no time to thicken long enough, before the morning sun already starts heating the ground and breaks the Fogs into patches and later maybe into fair-weather cumulus.

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